Technical Design Report for Subatomic Submarines Seaperch Team

ABSTRACT

This Technical Design Report (TDR) details how Subatomic Submarines developed a high-performance Remotely Operated Vehicle (ROV) to successfully complete the 2026 SeaPerch Mission and Obstacle Courses. Using the Engineering Design Process (EDP), the team iteratively designed, tested, and refined a submarine-style ROV optimized for speed, stability, and precision. The ROV features a modular, torpedo-inspired tubular design divided into three sealed sections. A clear resin hemispherical nose cone houses a forward-facing camera which reduces refraction and improves visibility for the pilot. The central section contains a system of custom Venturi tubes, including vertical and lateral configurations, which accelerate water flow to enhance thrust, maneuverability, and directional control. The team tested multiple Venturi tube geometries, lengths, and diameters to optimize propulsion efficiency, resulting in measurable performance improvements over standard configurations. The rear section houses a linear actuator used to dynamically adjust the center of mass and stabilizing fins to minimize rolling and maintain steady controlled motion. The onshore control system consists of two custom-made chips, an Arduino, and an Xbox controller; that enables us to have a more secure, controllable ROV. Extensive testing and data collection guided improvements in propulsion efficiency, buoyancy distribution, and handling under varying loads. This design directly addresses common challenges such as drag, instability, and payload management while maintaining a compact and robust structure. Overall, the ROV demonstrates a unique and innovative approach that balances advanced engineering concepts with practical performance, with potential applications in underwater inspection, maintenance, and defense systems.

TASK OVERVIEW

The 2026 SeaPerch challenge, consisting of two courses, focuses on Storm Response (Robonation, 2025). The timed Obstacle Course tests the speed of the ROV: It starts at the surface and wall, goes through five 18” hoops, placed at different angles, resurfaces, and comes back in reverse order. navigate through five hoops, surface, and retrace the route. The Mission Course tests the ROV’s ability to perform repairs and clear debris following a storm. The course involves four scored tasks (max. 110 pts) completed on the following set-up: a “surface vessel” for start and end operations; the front platform with bridge supports, marker floats, and a damaged pipe; and the back platform with the dam wall, debris field, and water sampler. Task #1 (max. 26 pts): Inspect the bridge by retrieving the red float and hooking it to the right rear pillar (10 pts); sliding the cover pipe to repair the support beam (8 pts total); and releasing the green float from the front left pillar (8 pts). Task #2 (max. 24 pts): Survey the dam by removing the plug from the surface vessel (4 pts), placing it into the hole on the dam wall (12 pts), and rotating the flood gate cover (8 pts). Task #3 (max. 38 pts): Clear the debris field by removing and relocating marine life (12 pts total); removing the heavy submerged debris (6 pts) and placing it onto a hook on the surface vessel (20 pts). Task #4 (max. 12 pts): Sample water quality by retrieving the water sampler (4 pts) and placing it onto a hook on the surface vessel (8 pts). Bonus points are awarded for completing all tasks in under 6 minutes (5 pts) and under 4 minutes (10 pts).

The described pool courses and tasks tremendously impacted the final design of our vehicle. Our ROV’s compact, stable, and lightweight structure enhances maneuverability and control across all platforms. Streamlined components and organized tether management reduce drag, improving speed during the obstacle course. A low profile and propeller walk helps mitigate unwanted forces and increase instant acceleration for both courses. A hook is specifically designed to complete all tasks in the mission course: it grabs and transports objects, lifts heavy debris and the dam plug, rotates the flood gate, and slides the pipe to repair the bridge structure.

DESIGN APPROACH

After analyzing the 2026 SeaPerch Mission and Obstacle Course requirements, our team established three primary objectives: maximize speed, improve maneuverability, and maintain stability during complex tasks. The obstacle course requires rapid navigation through angled hoops, while the mission course demands precise control and consistent orientation when manipulating objects. To achieve these goals, we applied the Engineering Design Process (EDP): **Ask**, **Imagine**, **Plan**, **Create**, **Test**, and **Improve** (Higdon, 2008) across all major subsystems. Our strategy emphasized data-driven iteration, where each design decision was validated through robust testing and refined based on measured performance.

Hydrodynamic Body and Camera System

Ask: How can we minimize drag while maintaining clear visibility?
Imagine: Flat fronts, angled wedges, and curved geometries with internal vs. external cameras
Plan: A streamlined tubular body with a hemispherical nose cone and internal camera
Create: A clear resin nose cone with a front-facing fishing camera
Test: Compared drag and visibility during timed trials
Improve: Adjusted camera placement to reduce distortion and obstruction


Figure 1. Hemispherical nose cone.

Drag force showed that reducing the drag coefficient and improving shape would significantly increase speed. The hemispherical nose cone (Figure 1) reduces pressure drag and flow separation, allowing smoother laminar flow around the ROV. Unlike traditional SeaPerch designs with exposed components, our enclosed camera system preserves the streamlined profile while improving pilot visibility. Testing confirmed improved navigation precision and reduced resistance compared to flat-front designs.

Venturi Tube Propulsion System

Ask: How can we increase thrust and propulsion efficiency beyond standard SeaPerch propellers?
Imagine: We explored open propellers, ducted systems, and Venturi-based propulsion. (Sharma, 2025)
Plan: Venturi tubes were selected due to their ability to accelerate water flow and improve thrust efficiency.
Create: Multiple prototypes were constructed with varying lengths, diameters, and taper angles.
Test: Each configuration was tested for speed, thrust output, and maneuverability,

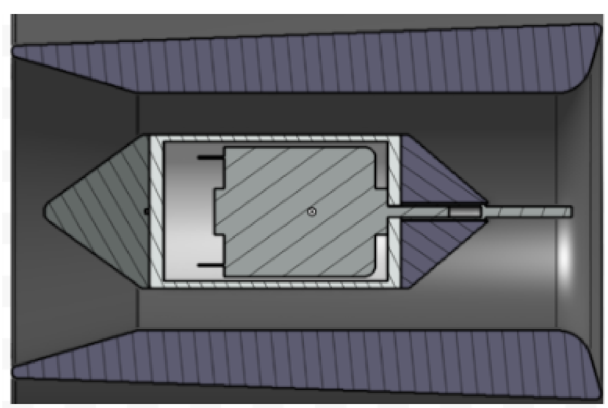
Figure 2. Cross-sectional geometry of the optimized Venturi tube.

revealing trade-offs between acceleration and flow losses.

Improve: The geometry was optimized to balance velocity increase with turbulence and friction.

Using the continuity equation and Bernoulli's principle, we determined that decreasing cross-sectional area increases velocity and reduces pressure. This increases thrust by accelerating water through the tube. Through robust testing, we identified an optimal configuration (Figure 2) that maximized exit velocity while minimizing turbulence. Both horizontal and vertical Venturi tubes were implemented for propulsion and depth control. This system replaces traditional open propellers with a fully integrated flow-accelerating propulsion system, significantly improving efficiency and control.

<u>Modular Structure and System Integration</u>

Ask: How can we design the ROV for easy modification, repair, and efficient system integration?

Imagine: We evaluated single-piece designs versus modular segmented systems.

Plan: A three-section modular design was selected to isolate key subsystems and allow rapid iteration.

Create: The ROV has front, central, and rear sections sealed using O-rings and secured with bolts (Figure 3).

Test: The modular system enabled quick disassembly during testing, allowing efficient adjustments to individual components.

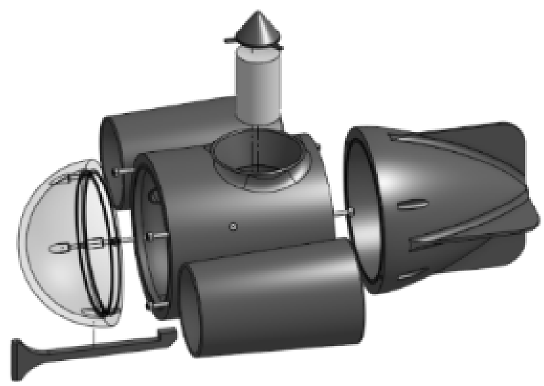

Figure 3. Three-section modular ROV design, exploded view.

Improve: Connections between sections were reinforced to improve durability and waterproofing.

The central section houses the Venturi propulsion system, positioning it for optimal water intake and flow acceleration. The front section contains the nose cone and camera system, while the rear section houses electronics, the linear actuator, and stabilization components. This modular design allowed each subsystem to be developed and tested independently while maintaining cohesion. It also improved the efficiency of the design process by enabling rapid iteration and targeted improvements.

By systematically applying the Engineering Design Process to each subsystem, our team developed a highly optimized ROV (Figure 4) that integrates hydrodynamic efficiency, advanced propulsion, and precise stability control. Each subsystem was independently refined through robust testing and iteration, then combined into a cohesive final design. The streamlined body reduces drag, the Venturi tube system maximizes thrust efficiency, and the combination of wheel weights and a linear actuator ensures stability and control. This structured, data-driven approach allowed us to move beyond traditional SeaPerch designs and create a unique, high-performance ROV. Through continuous testing, collaboration, and refinement,

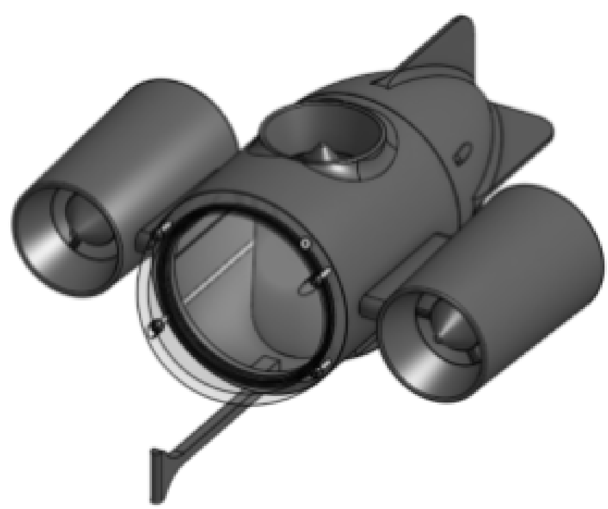

Figure 4. Final ROV Product.

we successfully engineered a system capable of excelling in both the Mission and Obstacle Courses.

EXPERIMENTAL RESULTS

We experimented with the modification of the geometries of our ROV in order to get the highest speed.
Experiment 1: In order to test the thrust of the ROV we suspended the Venturi tube in water and used a force sensor to measure the thrust of the ROV in Newtons. The venturi tubes help reduce blade drag and increase the efficiency of the propeller. We created 4 different venturi tubes and tested their thrust, along with a control. We hypothesised that E would be the best as it would allow for flow around the propellers due to the venturi effect. As shown in Figure 5, Venturi Tube D produced the highest average thrust of 4.5 N, outperforming Tube E by approximately 4.7% and significantly exceeding the baseline motor without a tube. The graph demonstrates a clear relationship between inlet geometry and thrust output, where smaller inlet angles reduce turbulence and improve flow alignment through the propeller. This results in more efficient momentum transfer from the motor to the water. The relatively small difference between Tubes D and E also suggests diminishing returns beyond a certain optimization point, indicating that further geometric refinement would yield only marginal gains. Therefore we decided to use Tube D design on the ROV.

Figure 5. Average thrust produced by different Venturi tube geometries.

Experiment 2: To determine the most efficient nose design for our ROV, we tested four shapes: cone, ogive, hemisphere, and flat. We hypothesized that the ogive or cone would perform best due to their streamlined profiles. Using OpenFoam CFD simulations under identical conditions, we compared drag coefficients and flow behavior across multiple trials. The flat nose had the highest drag (Cd ≈ 1.10), followed by the cone (Cd ≈ 0.50) and ogive (Cd ≈ 0.42). As illustrated in Figure 6, the hemispherical nose achieved the lowest drag coefficient (Cd ≈ 0.35), representing a reduction of approximately 16.7% compared to the ogive shape and over 68% compared to the flat design. The graph highlights a clear trend in which smoother curvature leads to reduced flow separation and improved pressure distribution. At the low Reynolds numbers typical of our ROV's operating conditions, maintaining attached flow due to the Coanda effect (Nuclear Energy, 2024) is more critical than minimizing frontal sharpness. This explains why the hemisphere outperformed sharper geometries, confirming that drag reduction in this regime is dominated by flow stability rather than purely aerodynamic streamlining. Although this contradicted

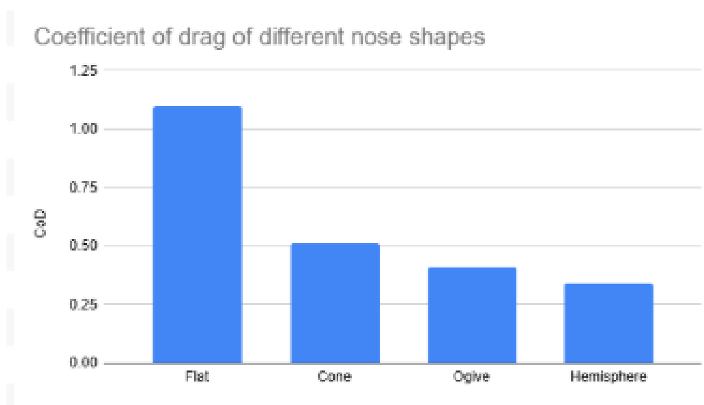

Figure 6. Drag coefficient comparison for different nose cone geometries.

our hypothesis, we selected the hemispherical design as it provided the best overall efficiency and stability.

REFLECTION & NEXT STEPS

By following the Engineering Design Process (EDP), our team overcame numerous challenges throughout the 2026 SeaPerch season and developed an ROV capable of effectively completing both the Mission and Obstacle Courses. The EDP was essential in guiding our decisions, particularly when balancing speed, stability, and precision—three design goals that often conflicted during early iterations. Through repeated testing and refinement, we learned that improving one aspect of performance, such as thrust, could negatively affect others like control or efficiency. This forced us to take a systems-level approach, where each modification was evaluated based on its overall impact on the ROV rather than isolated performance gains. Our modular, torpedo-inspired design proved to be one of our most successful decisions. Dividing the ROV into three sections allowed us to rapidly test, modify, and improve individual subsystems without rebuilding the entire vehicle. This significantly increased our efficiency during limited pool testing time. The streamlined body and hemispherical nose cone reduced drag and improved flow behavior, while also enhancing visibility through the internal camera system. One of the most important lessons we learned was the importance of data-driven decision-making. For example, although we initially predicted a different optimal Venturi tube design, experimental testing showed that Tube D produced the highest thrust. Similarly, CFD analysis revealed that a hemispherical nose minimized drag at our operating speeds, contradicting our original hypothesis. These experiences emphasized the importance of validating ideas through experimentation and being willing to adapt when results differ from expectations. We also gained valuable insight into system integration and real-world engineering trade-offs. While individual components such as the propulsion system, buoyancy control, and actuator performed well independently, ensuring they worked together seamlessly required additional refinement. Small adjustments in weight distribution, buoyancy, and propulsion alignment had significant effects on stability and maneuverability during complex tasks (SEAMOR, 2023). This reinforced the idea that successful engineering solutions depend on optimizing the entire system, not just individual parts. Throughout the season, our team developed strong collaboration, communication, and problem-solving skills. Long hours of building and testing allowed us to learn from failures, iterate quickly, and build on each other’s ideas. This collaborative environment was critical in refining our design and overcoming unexpected challenges.

Moving forward, we have identified several specific areas for improvement. First, we plan to further optimize the Venturi tube geometry by smoothing internal surfaces and refining inlet and outlet taper angles to reduce turbulence and maximize efficiency. Second, we aim to improve waterproofing and structural durability at modular joints to increase reliability during extended operation. Third, we want to enhance control precision by incorporating sensor feedback or partial automation, which could improve stability and reduce pilot workload during complex tasks. Additionally, we plan to improve tether management by reducing drag and minimizing interference, potentially through more streamlined cable designs or protective coatings. Beyond the ROV itself, we hope to mentor future teams and share our experience with iterative design, testing, and competition strategy. By continuing to refine both our engineering approach and teamwork, we believe we can further improve performance and contribute to future success in SeaPerch and similar engineering challenges.

ACKNOWLEDGEMENTS

This season would not have been possible without the contributions of others. We would like to acknowledge Mr. Brendan Wischweh for his expertise in physics and electronics. Furthermore, we extend our gratitude towards Mr. Peter Lambert and Waubonsie Valley High School for allowing us access to their pool for testing. We would also like to thank Mr. Ranga Mudireddy, our chaperone. We further acknowledge the donation of funds and material by Regression Robotics, a subsidiary of Dapper Ducks, NFP. We acknowledge the contributions of our logistics partner, Amtrak. Finally, we would like to thank our parents, peers, teachers, and others that have inspired us throughout the season.

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BUDGET

Material	Vendor	Unit Price (USD)	Quantity	Total Price
Fishing Camera	FOURQ	$33.29	1	$33.29
Arduino Uno	ELEGOO	$14.99	1	$14.99
Motor Controller	JLCPCB	$5.20	1	$5.20
USB Shield	Aliexpress	$6.49	1	$6.49
O-Rings	O-Ring Store	$0.36	3	$0.72
ABS for 3D print	Polymaker	$0.02/g	412.38 g	$8.24
Xbox controller	YAEYE	$15.98	1	$15.98
Resin for 3D print	PCBWay	$0.02/g	182.18 g	$3.64
Threaded Rods	Harfington	$0.84	4	$3.36
M3 Nuts	REV Robotics	$0.09	8	$0.72
Linear Actuator	Folany	$25.99	1	$25.99
Wheel Weights	ZNDAW	$0.13	48	$6.16

Supervisor Signature: Total Cost: $124.78